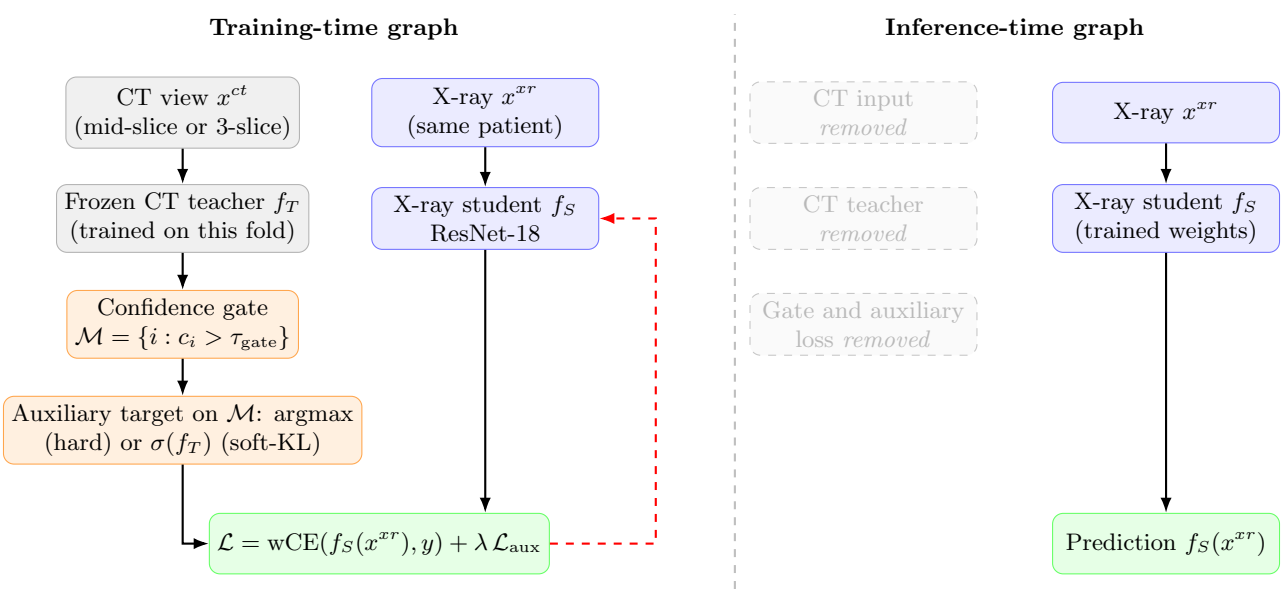

Training-time graph
CT view x^{ct} (mid-slice or 3-slice)
X-ray x^{xr} (same patient)
Frozen CT teacher f_T (trained on this fold)
X-ray student f_S ResNet-18
Confidence gate $\mathcal{M} = \{i : c_i > \tau_{\text{gate}}\}$
Auxiliary target on $\mathcal{M}$: argmax (hard) or $\sigma(f_T)$ (soft-KL)
$\mathcal{L} = \text{wCE}(f_S(x^{xr}), y) + \lambda\,\mathcal{L}_{\text{aux}}$
Inference-time graph
CT input removed
CT teacher removed
Gate and auxiliary loss removed
X-ray x^{xr}
X-ray student f_S (trained weights)
Prediction $f_S(x^{xr})$